5/1/2021

Lecture/Week (10)

LU-Factorization

Given:- Linear System

for example

$$3x_1 + 4x_2 - x_3 = 1$$
$$5x_1 + 3x_2 + 2x_3 = -1$$
$$-x_1 + x_2 - 3x_3 = 2$$

Methods

1) Dolittle Method
2) Crout's Method
3) Cholesky's Method

Dolittle Method

Step I :- Write linear system in matrix form

$$AX = b$$

Matrix of coefficient → A; known vector → b; Unknown vector → X

Step II :- Decompose matrix 'A' into lower triangular and upper triangular matrices. That is

$$A = LU$$

Step III :- Solve system

$LY = b$ for 'Y'

Step IV solve $UX = Y$ for 'X'.

Example 1

$$3x_1+4x_2-x_3=8$$
$$5x_1+3x_2+2x_3=17$$
$$-x_1+x_2-3x_3=-8$$

Solution

step I :

$$\underbrace{\begin{bmatrix}3 & 4 & -1\\5 & 3 & 2\\-1 & 1 & -3\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}}_{X}=\underbrace{\begin{bmatrix}8\\+17\\-8\end{bmatrix}}_{b}$$

step II :-

$$A = L\,U$$

$$\begin{bmatrix}3 & 4 & -1\\5 & 3 & 2\\-1 & 1 & -3\end{bmatrix}=\begin{bmatrix}l_{11} & 0 & 0\\l_{21} & l_{22} & 0\\l_{31} & l_{32} & l_{33}\end{bmatrix}\begin{bmatrix}U_{11} & U_{12} & U_{13}\\0 & U_{22} & U_{23}\\0 & 0 & U_{33}\end{bmatrix}$$

Dolittle Method :- Take the diagonal elements of matrix 'L' equal to 1's.

Crout's Method :- Take the diagonal elements of matrix 'U' equal to 1's.

Dolittle Method

$$\begin{bmatrix}3 & 4 & -1\\5 & 3 & 2\\-1 & 1 & -3\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\l_{21} & 1 & 0\\l_{31} & l_{32} & 1\end{bmatrix}\begin{bmatrix}U_{11} & U_{12} & U_{13}\\0 & U_{22} & U_{23}\\0 & 0 & U_{33}\end{bmatrix}$$

$$\begin{bmatrix}3 & 4 & -1\\5 & 3 & 2\\-1 & 1 & -3\end{bmatrix}=\begin{bmatrix}U_{11} & U_{12} & U_{13}\\l_{21}U_{11} & l_{21}U_{12}+U_{22} & l_{21}U_{13}+U_{23}\\l_{31}U_{11} & l_{31}U_{12}+l_{32}U_{22} & l_{31}U_{13}+l_{32}U_{23}+U_{33}\end{bmatrix}$$

Equating both sides

$\boxed{U_{11} = 3}$ $\boxed{U_{12} = 4}$ $\boxed{U_{13} = -1}$

$l_{21}U_{11} = 5$

$\boxed{l_{21} = \frac{5}{U_{11}} = \frac{5}{3}}$

$l_{21}U_{12} + U_{22} = 3$

$\frac{5}{3}(4) + U_{22} = 3$

$\boxed{U_{22} = \frac{-11}{3}}$

$l_{21}U_{13} + U_{23} = 2$

$\frac{5}{3}(-1) + U_{23} = 2$

$\boxed{U_{23} = \frac{11}{3}}$

$l_{31}U_{11} = -1$

$\boxed{l_{31} = \frac{-1}{U_{11}} = \frac{-1}{3}}$

$l_{31}U_{12} + l_{32}U_{22} = 1$

$-\frac{1}{3}(4) + l_{32}\left(\frac{-11}{3}\right) = 1$

$\boxed{l_{32} = \frac{-7}{11}}$

$l_{31}U_{13} + l_{32}U_{23} + U_{33} = -3$

$\left(\frac{-1}{3}\right)(-1) + \left(\frac{-7}{11}\right)\left(\frac{11}{3}\right) + U_{33} = -3$

$\boxed{U_{33} = -1}$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 5/3 & 1 & 0 \\ -1/3 & -7/11 & 1 \end{bmatrix} \qquad U = \begin{bmatrix} 3 & 4 & -1 \\ 0 & -11/3 & 11/3 \\ 0 & 0 & -1 \end{bmatrix}$$

Step III :-

$$LY = b$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 5/3 & 1 & 0 \\ -1/3 & -7/3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 17 \\ -8 \end{bmatrix}$$

$y_1 = 8 \longrightarrow \boxed{y_1 = 8}$

$\frac{5}{3}y_1 + y_2 = 17 \rightarrow \frac{5}{3}(8) + y_2 = 17 \quad \boxed{y_2 = \frac{11}{3}}$

$-\frac{1}{3}y_1 - \frac{7}{3}y_2 + y_3 = -8 \rightarrow -\frac{1}{3}(8) - \frac{7}{3}\left(\frac{11}{3}\right) + y_3 = -8$

$\boxed{y_3 = -3}$

Step IV :- $UX = Y$

$$\begin{bmatrix} 3 & 4 & -1 \\ 0 & -11/3 & 11/3 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 11/3 \\ -3 \end{bmatrix}$$

$$3x_1 + 4x_2 - x_3 = 8 \rightarrow 3x_1 + 4(2) - (3) = 8 \rightarrow \boxed{x_1 = 1}$$

$$\frac{-11}{3} x_2 + \frac{11}{3} x_3 = 11/3 \rightarrow \frac{-11}{3} x_2 + \frac{11}{3}(3) = \frac{11}{3} \quad \boxed{x_2 = 2}$$

$$-x_3 = -3 \longrightarrow \boxed{x_3 = 3}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$ Ans.

Example 2 — Crout's Method

$$\begin{bmatrix} 3 & 4 & -1 \\ 5 & 3 & 2 \\ -1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 17 \\ -8 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 & -1 \\ 5 & 3 & 2 \\ -1 & 1 & -3 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 & -1 \\ 5 & 3 & 2 \\ -1 & 1 & -3 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}U_{12} & l_{11}U_{13} \\ l_{21} & l_{21}U_{12}+l_{22} & l_{21}U_{13}+l_{22}U_{23} \\ l_{31} & l_{31}U_{12}+l_{32} & l_{31}U_{13}+l_{32}U_{23}+l_{33} \end{bmatrix}$$

$\boxed{l_{11} = 3}$

$\boxed{l_{21} = 5}$

$\boxed{l_{31} = -1}$

$l_{11}U_{12} = 4$

$3U_{12} = 4$

$\boxed{U_{12} = \frac{4}{3}}$

$l_{11}U_{13} = -1$

$3U_{13} = -1$

$\boxed{U_{13} = \frac{-1}{3}}$

$l_{21}U_{12} + l_{22} = 3$

$5\left(\frac{4}{3}\right) + l_{22} = 3$

$\boxed{l_{22} = \frac{-11}{3}}$

$\cancel{l_{31}U_{13} + l_{32}U_{23} + l_{33} = -3}$

$l_{21}U_{13} + l_{22}U_{23} = 2$

$5\left(\frac{-1}{3}\right) + \left(\frac{-11}{3}\right)U_{23} = 2$

$\boxed{U_{23} = \frac{22}{3}}$

$l_{31}U_{12} + l_{32} = 1$

$(-1)\left(\frac{4}{3}\right) + l_{32} = 1$

$l_{32} = 7/3$

$l_{31}U_{13} + l_{32}U_{23} + l_{33} = -3$

$(-1)\left(\frac{-1}{3}\right) + \left(\frac{7}{3}\right)\left(\frac{22}{3}\right) + l_{33} = -3$

$\boxed{l_{33} = -}$

Step III $\quad LY = b$

Step IV $\quad Ux = Y$